DOCUMENT 5:
DTAF AI PIPELINE QUICK DEMO

This guide is intended to be used for a quick demonstration of the DTAF AI Pipeline. For an in-depth explanation of the DTAF AI Pipeline application, the user should reference the "Document 3: DTAF AI PIPELINE- CNC CASE STUDY" document. The remainder of the document is structured according to the following: Section 1 provides the user inputs for DTAF initialization, Section 2 provides the user inputs for the feasibility check, and Section 3 shows which documents the user needs to download to support the primary file.

1. Initialization

- Respond to "What sector should be analyzed" (Figure 1)?
    - Enter **"Manufacturing".**
    - Press **"Enter".**

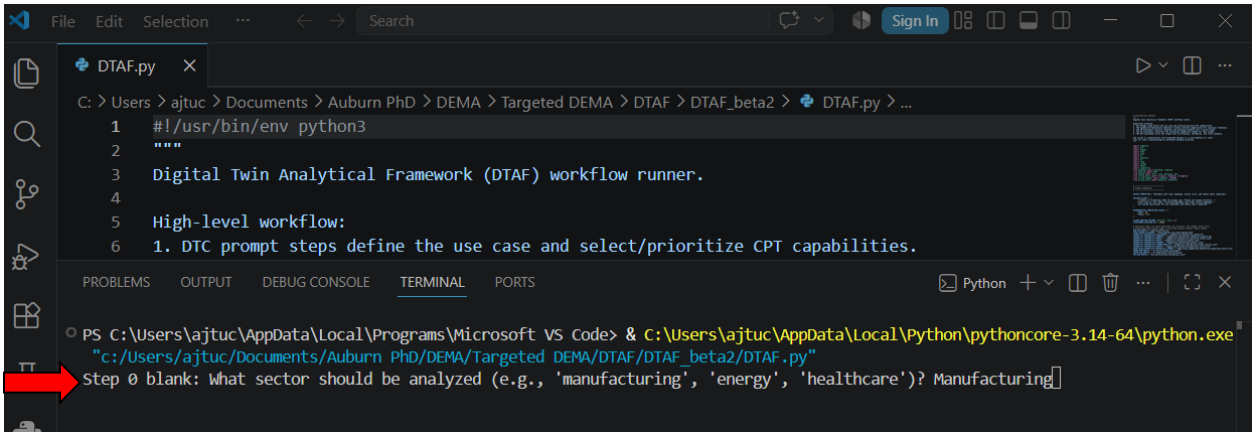


Figure 1: Visual Studio Code for Sector Selection.

- Enter the following problem statement:
    - *"We need a digital twin of our CNC machining systems to provide accurate, data-driven insight into Quality, Cost, and Time before production begins. The digital twin must determine whether we have the capability and capacity to run a given job, optimize scheduling based on available equipment, and automate key decision-making processes that currently rely on manual judgment. It should continuously monitor the degradation of cutting tools, spindles, and coolant, predict remaining useful life, and assess how component wear will impact part quality and throughput. By integrating real-time machine data with predictive simulation, the digital twin will enable proactive maintenance, reduce cycle time, extend asset life, and ensure that every job is run on the right machine, at the right time, with predictable and efficient performance."*
- Press **Shift + Enter** to move to a new line. Type "**END"** in capital letters, and then press "**Enter"** (Figure 2).

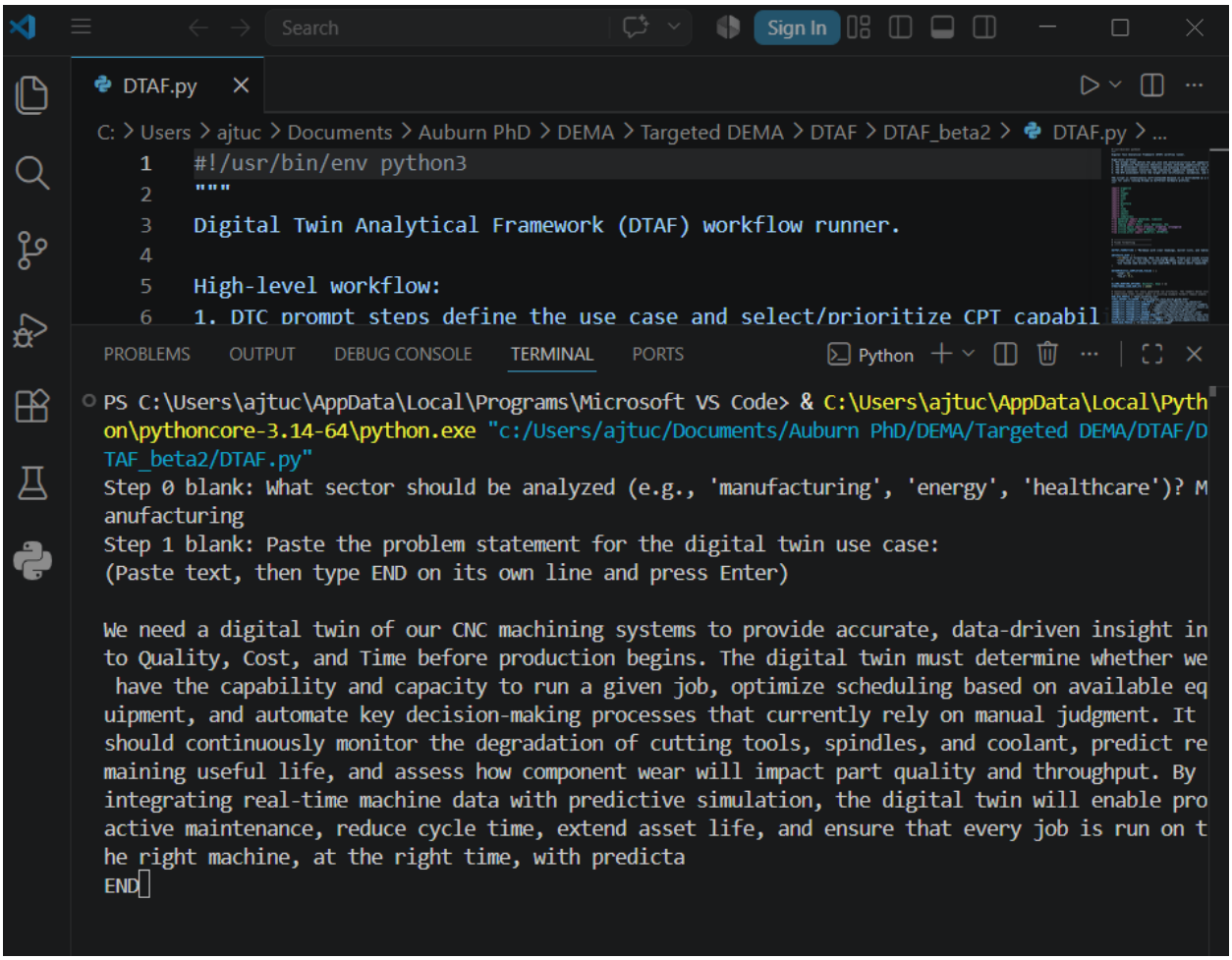


Figure 2: Visual Studio Code for Problem Statement.

2. Feasibility check

Choose one pathway: Feasibility Checklist or DEMA

- Complete the Feasibility Checklist (Figure 3):
  - The pipeline will present several questions associated with the Essential capabilities.
  - Answer **"ALL"** for each of the questions.
  - Press **Enter**.
  - This input is used only to reproduce the predefined demonstration and should not be treated as an actual readiness assessment.

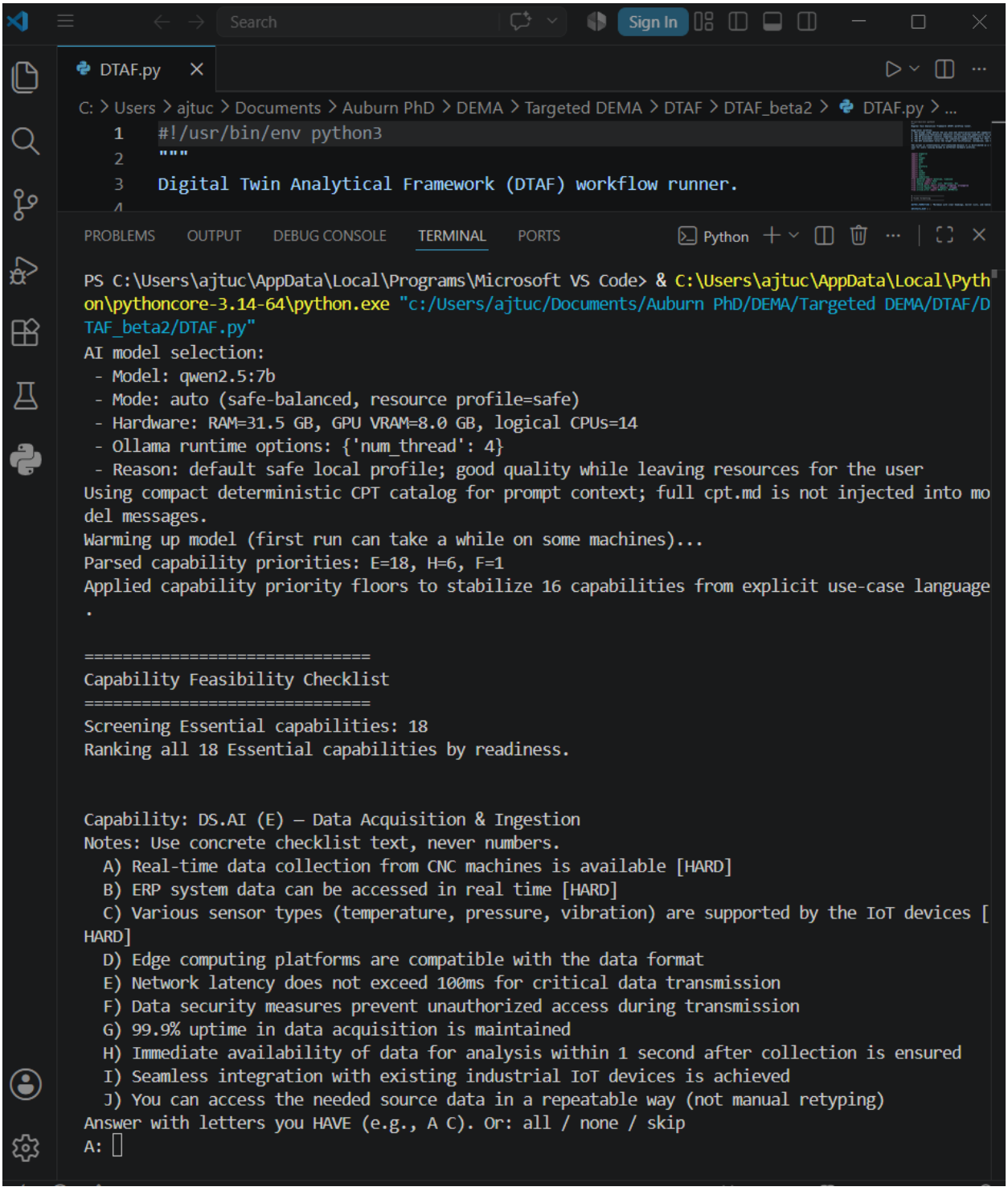


Figure 3: Visual Studio Code for Feasibility Checklist Questions .

- For DEMA Option:
    - Download the file “DEMA_CNC_DATA_ELEMENT_LEVEL_VIEW.xlsx” from the DEMA Software that is installed on the MxD computer (Figure 4). Email anb0051@auburn.edu for the document if you are not an MxD member.
        a. Note when opening the software that the DEMA results for this particular demo have already been input into the software. If someone were to be starting a new project, they would need to create a new project in the software. The accompanying “DEMA CNC CASE STUDY” document shows how to create a new data set in the DEMA Software.

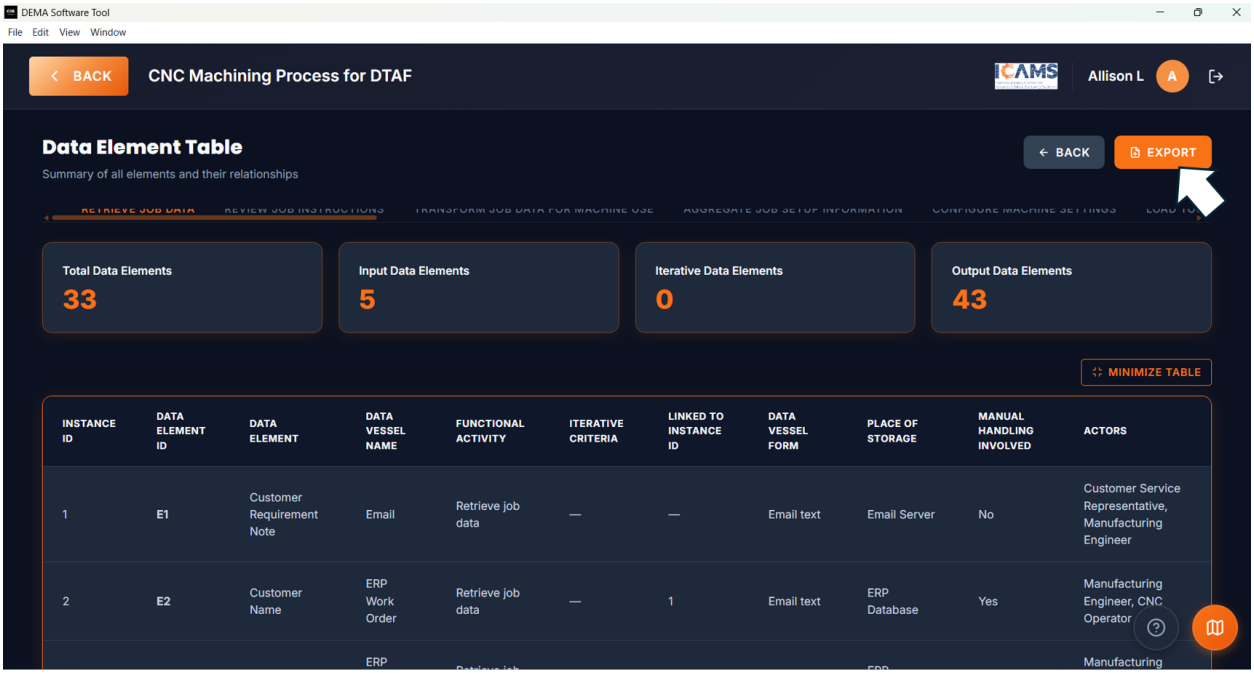


Figure 4: Download the DEMA Data Set.

3. Download the Following Files

- Primary file: DT Build Guide - open this file for a holistic overview of the DT guide.
    - dtaf_workflow_run_20260702_154927\dtaf_digital_twin_build_guide.html
- Supporting Files - Download these files for more details on specific outputs.
    - step0_use_case_suitability.md
    - step1_business_requirements.md
    - step2_capability_selector.md
    - step3_capability_deep_dives.md
    - priorities.json
    - step4_cpt_table.html
    - Run_log.jsonl
- Feasibility Supporting File
    - Capability_feasibility_checklist_20260702_155510\capability_feasibility_checklist_summary.md
- 4R Supporting Files

    - 4r_build_target_assessment_20260702_162452\4r_assessment_summary.md
    - 4r_build_target_assessment_20260702_162452\4r_assessment_result.json
- 4S Supporting files
    - 4s_simulation_classification_20260702_162619\4s_assessment_summary.md
    - 4s_simulation_classification_20260702_162619\4s_assessment_result.json
- DTV Supporting Files
    - dtv_verification_validation_trust_20260702_162719\dtv_assessment_summary.md
    - dtv_verification_validation_trust_20260702_162719\dtv_assessment_result.json

4. Explaining the Results

- Opening the Build Guide
    - The Build Guide is the primary output of the DTAF AI Pipeline. It combines the use case analysis, capability recommendations, feasibility results, Digital Twin maturity assessment, simulation guidance, readiness gaps, build roadmap, and verification and validation guidance into one report (Figure 5). The supporting files preserve the detailed analysis used to create the summary.

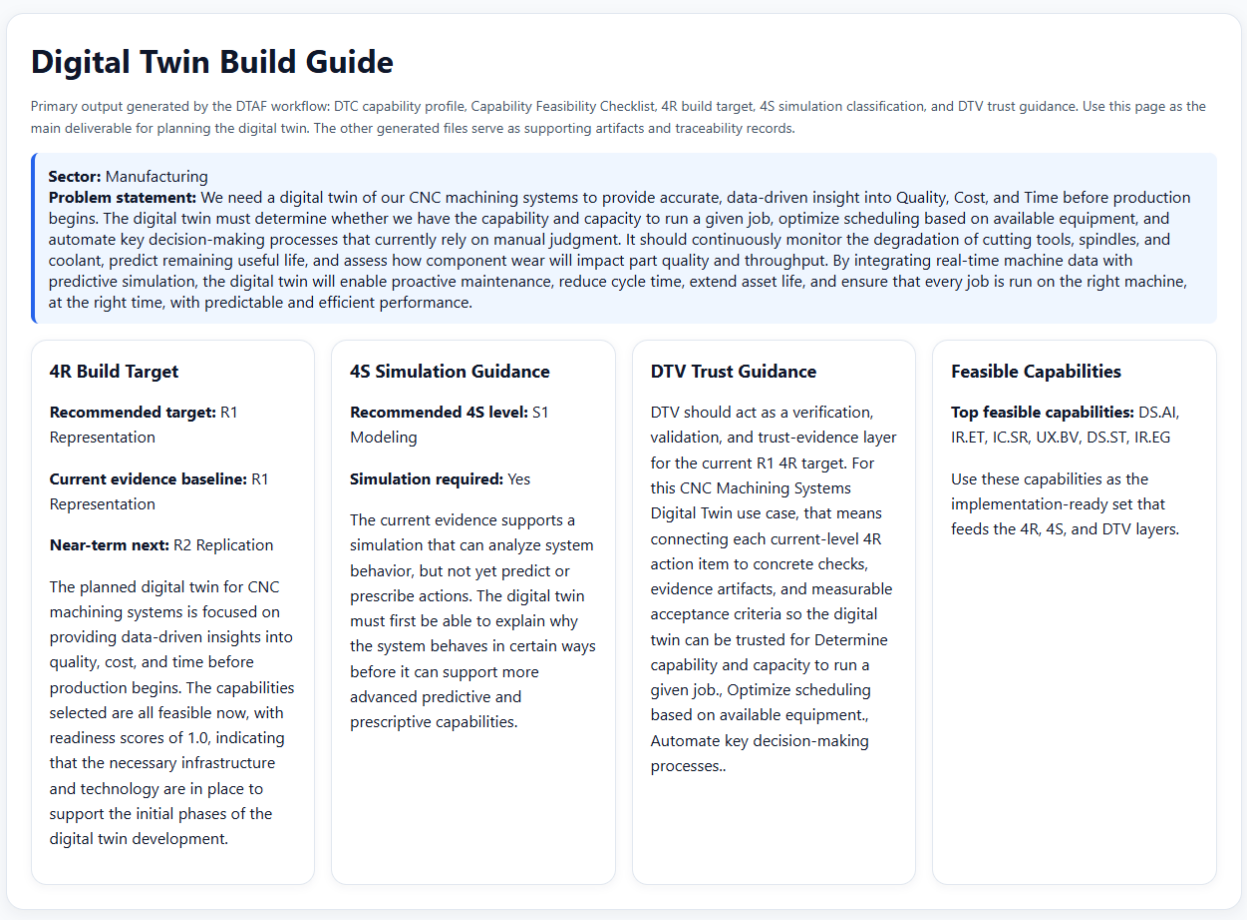


Figure 5: Digital Twin Build Guide for the Feasibility Checklist Demo.

- Explaining the Digital Twin Capabilities Periodic Table
  - This periodic table presents the capabilities for the DT in several categories of priority. The Essential category is the one that the DTAF pipeline uses to ask the user the feasibility questions (Figure 6).

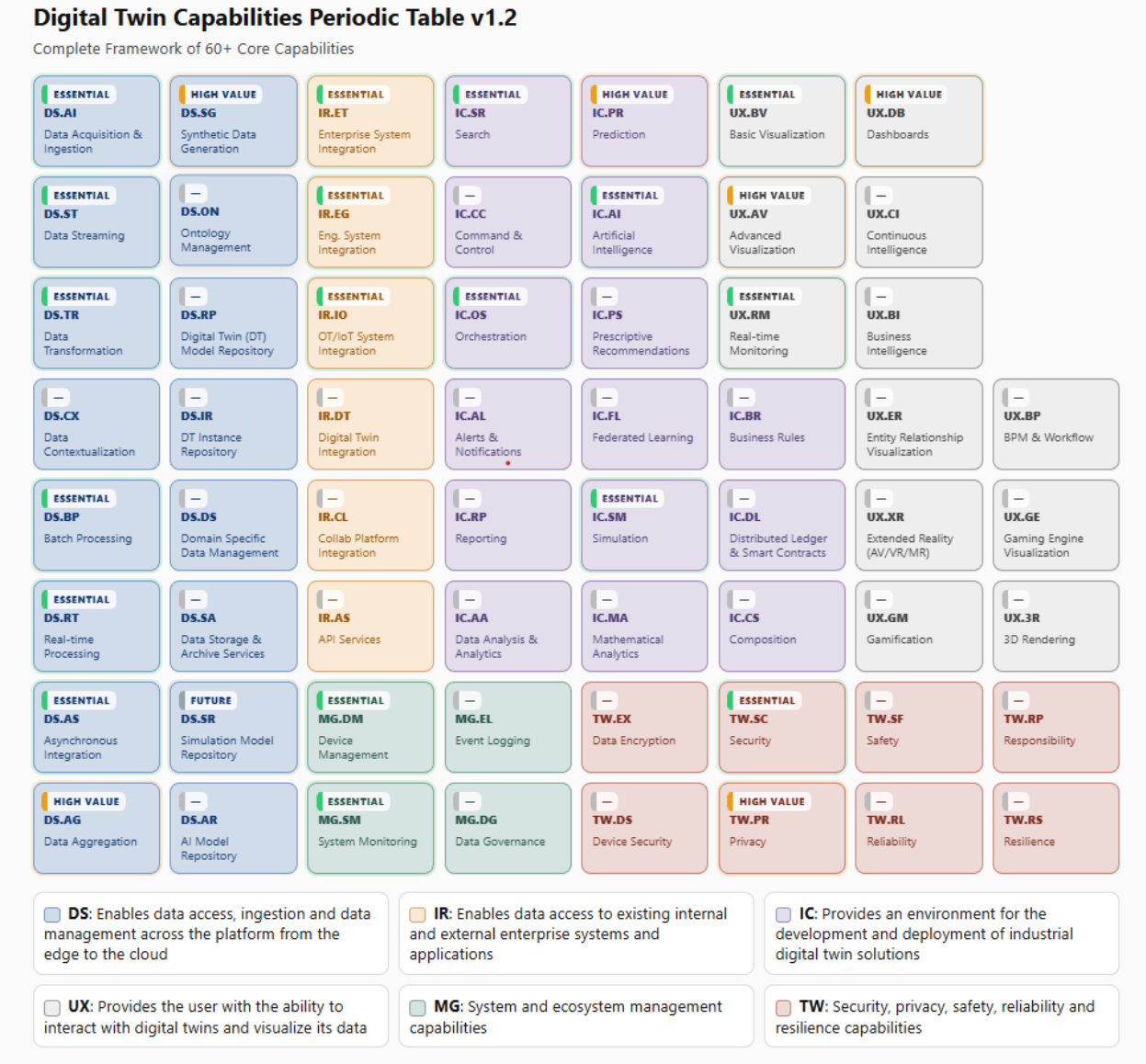


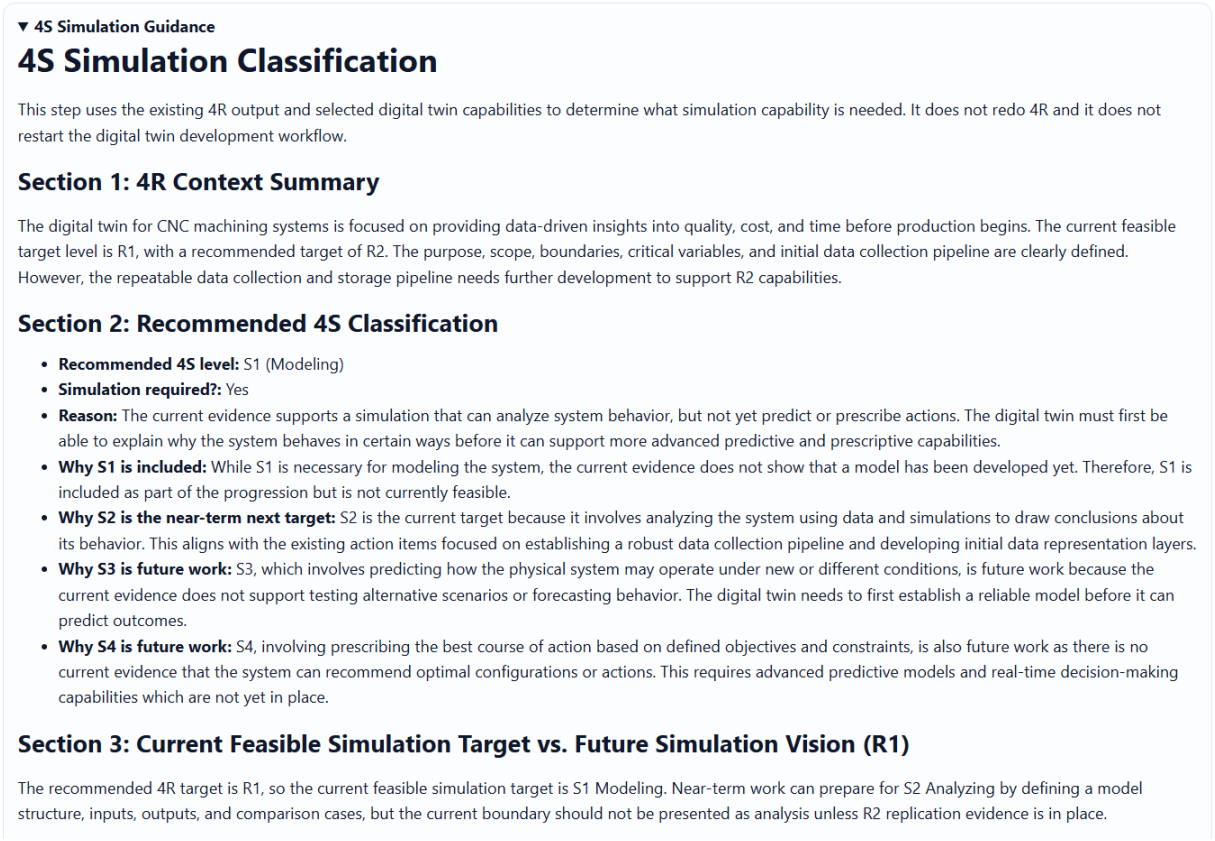
Figure 6: Capabilities Periodic Table for the Feasibility Checklist Demo.

- Explaining the 4R Results
  - The recommended Digital Twin build target is R2, or Replication, while the current evidence baseline is R1, or Representation. This means the available information supports creating a structured digital representation of the CNC system, but the system needs additional development before it can reliably replicate the behavior of the physical CNC process in a virtual environment. The build roadmap focuses on closing the gap between R1 and R2 rather than immediately pursuing levels of R3 and R4 (Figure 7).



Figure 7: Expanded 4R Results for the Feasibility Checklist Demo.

- Explaining the 4S Result
    - The recommended simulation level is S2, or Analyzing, and the report indicates that simulation is required. The Digital twin must do more than display the CNC system. It must model and analyze machine behavior, production conditions, maintenance, quality, and throughput to provide data-driven insight. The result does not yet require the simulation to prescribe or autonomously execute every decision (Figure 8).



▼ 4S Simulation Guidance

4S Simulation Classification

This step uses the existing 4R output and selected digital twin capabilities to determine what simulation capability is needed. It does not redo 4R and it does not restart the digital twin development workflow.

Section 1: 4R Context Summary

The digital twin for CNC machining systems is focused on providing data-driven insights into quality, cost, and time before production begins. The current feasible target level is R1, with a recommended target of R2. The purpose, scope, boundaries, critical variables, and initial data collection pipeline are clearly defined. However, the repeatable data collection and storage pipeline needs further development to support R2 capabilities.

Section 2: Recommended 4S Classification

- **Recommended 4S level:** S1 (Modeling)
- **Simulation required?:** Yes
- **Reason:** The current evidence supports a simulation that can analyze system behavior, but not yet predict or prescribe actions. The digital twin must first be able to explain why the system behaves in certain ways before it can support more advanced predictive and prescriptive capabilities.
- **Why S1 is included:** While S1 is necessary for modeling the system, the current evidence does not show that a model has been developed yet. Therefore, S1 is included as part of the progression but is not currently feasible.
- **Why S2 is the near-term next target:** S2 is the current target because it involves analyzing the system using data and simulations to draw conclusions about its behavior. This aligns with the existing action items focused on establishing a robust data collection pipeline and developing initial data representation layers.
- **Why S3 is future work:** S3, which involves predicting how the physical system may operate under new or different conditions, is future work because the current evidence does not support testing alternative scenarios or forecasting behavior. The digital twin needs to first establish a reliable model before it can predict outcomes.
- **Why S4 is future work:** S4, involving prescribing the best course of action based on defined objectives and constraints, is also future work as there is no current evidence that the system can recommend optimal configurations or actions. This requires advanced predictive models and real-time decision-making capabilities which are not yet in place.

Section 3: Current Feasible Simulation Target vs. Future Simulation Vision (R1)

The recommended 4R target is R1, so the current feasible simulation target is S1 Modeling. Near-term work can prepare for S2 Analyzing by defining a model structure, inputs, outputs, and comparison cases, but the current boundary should not be presented as analysis unless R2 replication evidence is in place.

Figure 8: Expanded 4S Results for the Feasibility Checklist Demo.

- Explaining the DTV Results
    - The DTV section explains how trust will be established in the R2 Digital Twin. It connects the recommended build actions to verification checks, validation activities, evidence requirements, and measurable acceptance criteria. For this CNC case, the evidence may include machine sensor data, maintenance logs, quality control records, production schedules, and historical job data. The purpose is to confirm that the Digital Twin was built correctly and that its outputs accurately represent the physical CNC system well enough to support the intended decisions (Figure 9).

▼ DTV Verification and Validation Guidance

Digital Twin V (DTV) Guidance

This step uses the existing 4R output to explain how the proposed digital twin should be verified, validated, and trusted for the specific use case. It does not redo 4R and it does not replace the existing 4R development path.

Section 1: 4R Context Summary

The recommended pre-build 4R target is R1 (Representation). The current evidence baseline is R1 (Representation). The near-term next target beyond the current recommendation is R2 (Replication). The planned digital twin for CNC machining systems is focused on providing data-driven insights into quality, cost, and time before production begins. The capabilities selected are all feasible now, with readiness scores of 1.0, indicating that the necessary infrastructure and technology are in place to support the initial phases of the digital twin development.

Section 2: DTV Role for This Use Case

DTV should act as a verification, validation, and trust-evidence layer for the current R1 4R target. For this CNC Machining Systems Digital Twin use case, that means connecting each current-level 4R action item to concrete checks, evidence artifacts, and measurable acceptance criteria so the digital twin can be trusted for Determine capability and capacity to run a given job., Optimize scheduling based on available equipment., Automate key decision-making processes..

Section 3: Current R1 Trust Boundary

The current DTV focus is limited to representation trust evidence: whether the selected variables, data sources, schemas, metadata, and initial representations are implemented correctly and are trustworthy enough for the intended use. It should not yet claim replication, prediction, or autonomy evidence.

Figure 9: Expanded 4S Results for the Feasibility Checklist Demo.